

Stage 9 End of unit 2

The tests and mark schemes have been written by the authors. These may not fully reflect the approach of Cambridge Assessment International Education.

Answers

- 1** **a** 45 [2]
 b 0 [2]
 c 72 [2]

- 2** **a** $6x + 2$ [2]
 b $2x^2 + x$ [2]

- 3** **a** p^{12} [1]
 b q^6 [1]
 c r^{15} [1]
 d $6t^5$ [1]

- 4** **a** **B** $12x^5$ [1]
 b **C** $10y^6$ [1]
 c **A** $\frac{1}{2}k^8$ [1]
 d **C** $\frac{2}{3}m^7$ [1]
 e **D** $\frac{3}{2}n^{-4}$ [1]

- 5** **a** $y^2 + 8y + 15$ [1]
 b $m^2 - m - 2$ [1]
 c $p^2 - 8p + 15$ [1]
 d $n^2 - 16n + 64$ [1]
 e $x^2 - 4$ [1]

- 6** **a** $\frac{3x}{5}$ [1]
 b $\frac{y}{6}$ [1]

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- c** $\frac{2a+b}{8}$ [1]
- d** $\frac{4a+3b}{18}$ [1]
- e** $3x+9$ [1]
- 7 a** $T=30$ [2]
- b i** $p = \frac{T}{x}$ [1]
- ii** $p = 6$ [2]
- 8 a B** $x = \frac{y-c}{3}$ [1]
- b C** $x = 4(y+2k)$ [1]
- c A** $x = 4ay - b$ [1]
- 9 a** $A = 4x^2 + yz$ [2]
- b** $A = 64$ [2]
- c** $x = \sqrt{\frac{A-yz}{4}}$ [2]
- d** $x = 4$ [2]

[Total: 45 marks]